

A Toy Model for Protein Evolution

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Context: Proteins are highly complex molecules, vital to all cellular processes. They evolve through mutations while maintaining their biological functions. Even minor mutations can destabilize a protein, making their evolution a complex process to study.

The lattice protein model (LP) is a toy model that aims to simplify this complex process: a protein is represented as a chain folded on a square or cubic lattice, and interaction between neighbouring amino acids determine the protein stability [1,2]. This model captures the essence of protein folding and evolution while being simple enough for numerical simulation.

Project: implement a lattice protein model and simulate its evolution on a phylogenetic tree. This model will help benchmark phylogenetic methods, such as ancestral sequence reconstruction (ASR). Specifically, we aim to determine if ASR can reliably infer changes in specificity on a tree and if ancestral proteins were more thermostable [3].

We plan to:

1. Implement a simple LP model to generate amino acid sequences folding into a given structure.
2. Simulate the evolution of this protein on a tree.
3. Use the model to study ancestral sequence reconstruction techniques.

The thesis will focus on the initial steps, with further steps explored based on time and interest.

Expected skills & interests:

- Working with probabilistic/computational models.
- Learning a new programming language: we aim to write the code using the Julia language, which is both easy to prototype and fast.

- [1] E. Shakhnovich and A. Gutin, Enumeration of All Compact Conformations of Copolymers with Random Sequence of Links, *The Journal of Chemical Physics* **93**, 5967 (1990).
- [2] E. Loffredo, E. Vesconi, R. Razban, O. Peleg, E. Shakhnovich, S. Cocco, and R. Monasson, Evolutionary Dynamics of a Lattice Dimer: A Toy Model for Stability vs. Affinity Trade-Offs in Proteins, *Journal of Physics A: Mathematical and Theoretical* **56**, 455002 (2023).
- [3] L. C. Wheeler, S. A. Lim, S. Marqusee, and M. J. Harms, The Thermostability and Specificity of Ancient Proteins, *Current Opinion in Structural Biology* **38**, 37 (2016).